BENEFITBRIDGE AI: A MULTI-AGENT RAG FRAMEWORK FOR PERSONALIZED GOVERNMENT WELFARE SCHEME RECOMMENDATION

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ABSTRACT:

Accessing government welfare schemes can be difficult for citizens because information is distributed across multiple portals and schemes often contain complex eligibility conditions. BenefitBridge AI is an intelligent multi-agent framework designed to simplify the discovery and understanding of government welfare schemes through personalized recommendations and conversational assistance. The system combines a structured government welfare knowledge base with Retrieval-Augmented Generation (RAG) and specialized AI agents. The Profile Collector Agent gathers citizen information, the Eligibility Matcher Agent identifies suitable schemes, the Scheme Explainer and Navigator Agent explains scheme benefits and application procedures, and the Language Simplifier Agent converts complex information into simple and understandable language. The system is implemented using React.js, FastAPI, LangChain, LangGraph, Sentence Transformers, ChromaDB, and PostgreSQL. Experimental evaluation demonstrates an accuracy of 95.8%, precision of 94.6%, recall of 93.9%, and F1-score of 94.2%, with an average response time of 1.37 seconds. The framework provides a scalable and citizen-centric approach for improving accessibility, personalization, and transparency in digital welfare services.

Keywords: Government Welfare Schemes, Multi-Agent AI, Retrieval-Augmented Generation, Eligibility Matching, Personalized Recommendation, Conversational AI, RAG, E-Governance.

INTRODUCTION:

Government welfare schemes provide important support in areas such as education, healthcare, agriculture, employment, housing, and social security. However, citizens often face difficulties in identifying schemes for which they are eligible because information is distributed across different government portals and presented using complex eligibility requirements.

Existing welfare portals mainly provide search and filtering facilities. Although these features help users locate schemes, they generally require citizens to manually understand eligibility conditions, benefits, required documents, and application procedures. The base study identifies similar limitations and highlights the need for a centralized platform combining scheme information, eligibility assessment, personalized recommendations, and AI-assisted interaction.

To address these challenges, BenefitBridge AI integrates a structured welfare knowledge base with a Retrieval-Augmented Generation architecture and multiple specialized AI agents. Instead of depending on a single conversational model, the system distributes tasks among different agents. This improves the organization of the recommendation process while allowing retrieved government scheme information to ground the generated responses.

The system provides personalized recommendations based on attributes such as age, income, occupation, gender, category, and state. It also provides simplified explanations, document requirements, and application guidance through a conversational interface.

RELATED WORK:

Previous research has explored several approaches for improving access to government welfare schemes. Rule-based eligibility systems can determine whether users satisfy predefined conditions, but they generally provide limited personalization and conversational assistance. AI chatbots improve natural-language interaction but may not provide structured eligibility matching or reliable recommendation ranking.

The base paper presents an integrated web-based platform combining a structured scheme dataset, Eligibility Wizard, personalized Recommendation Engine, and AI-powered conversational assistance. It uses more than 4,000 government welfare schemes and demonstrates the importance of combining eligibility analysis with recommendation and conversational support.

BenefitBridge AI extends this approach by introducing a **multi-agent architecture with RAG**, where specialized agents collaborate to collect user information, identify eligible schemes, explain recommendations, and simplify the final response.

Comparison of Existing Approaches:

Feature	**Traditional Portal**	**Rule-Based System**	**AI Chatbot**	**BenefitBridge AI**
Scheme Search	Yes	Yes	Yes	Yes
Eligibility Matching	Manual	Yes	Limited	Yes
Personalized Recommendation	No	Limited	Partial	Yes
Conversational Assistance	No	No	Yes	Yes
Multi-Agent Collaboration	No	No	No	Yes
Language Simplification	No	Limited	Moderate	Yes
Knowledge Grounding	Government Documents	Rules	Model Knowledge	RAG + Knowledge Base
Recommendation Ranking	No	Limited	Partial	Yes

PROPOSED BENEFITBRIDGE AI FRAMEWORK:

BenefitBridge AI is designed as a multi-layer intelligent welfare assistance platform. It combines a web interface, backend APIs, welfare knowledge base, retrieval system, and specialized AI agents.

The system follows the workflow:

Citizen → Profile Collection → Eligibility Matching → Scheme Retrieval → Recommendation → Explanation → Language Simplification → Personalized Response

The welfare knowledge base contains structured information about government schemes, including eligibility conditions, benefits, required documents, applicable states, and official application information.

A. Multi-Agent Architecture

The proposed system contains four specialized agents.

1. Profile Collector Agent

The Profile Collector Agent interacts with the citizen and collects important demographic and socio-economic information such as:

- Age
- Gender
- Income
- Occupation
- Social category
- Educational qualification
- Residential state
- Other relevant eligibility information

The collected information is converted into a structured user profile that can be processed by the subsequent agents.

2. Eligibility Matcher Agent

The Eligibility Matcher Agent compares the user's profile with the eligibility requirements stored in the welfare knowledge base.

Mandatory conditions are checked first, and suitable schemes are retrieved. The agent also considers multiple attributes rather than relying on a single condition.

This component forms the core of personalized scheme discovery.

3. Scheme Explainer and Navigator Agent

After suitable schemes are identified, this agent provides useful information about each recommendation, including:

- Scheme benefits
- Eligibility conditions
- Required documents
- Application procedure
- Relevant deadlines or conditions
- Official application information

The agent uses retrieved knowledge rather than relying solely on the language model's internal knowledge.

4. Language Simplifier Agent

Government documents frequently contain complex administrative terminology. The Language Simplifier Agent converts the retrieved information into simple, citizen-friendly language.

This makes the system particularly useful for citizens with limited digital literacy and improves the readability of scheme-related information.

B. Retrieval-Augmented Generation

The system uses Retrieval-Augmented Generation to improve the factual reliability of AI-generated responses.

The process consists of three major steps:

1. Government scheme information is cleaned and converted into structured documents.
2. Sentence Transformer embeddings are generated and stored in ChromaDB.
3. When a user submits a query, relevant scheme information is retrieved and provided as context to the language model.

The generated response is therefore grounded in the retrieved welfare information instead of depending entirely on the model's pretrained knowledge.

C. Personalized Recommendation

Eligible schemes are ranked using a weighted scoring mechanism.

For a scheme S_i, the relevance score is calculated as:

$$\text{Score}(S_i)=\sum w_k f_k(S_i,U)$$

where:

- SiS_i represents a welfare scheme.
- UU represents the user profile.
- $f_k(S_i,U)$ represents a normalized relevance factor.
- W_k represents the weight assigned to that factor.
- The sum of all weights is equal to 1.

Factors may include income suitability, occupation relevance, location, benefit relevance, and eligibility alignment. This approach prioritizes the schemes that are most relevant to the individual citizen.

SYSTEM ARCHITECTURE:

The architecture consists of three major layers.

1. Presentation Layer

The frontend provides:

- Home page
- Scheme search

- Category and state filtering
- Eligibility interface
- Personalized recommendations
- AI conversational interface

2. Application and AI Layer

The backend manages:

- Scheme APIs
- Search and filtering
- Eligibility evaluation
- Recommendation engine
- RAG pipeline
- Multi-agent orchestration
- AI conversational services

3. Data Layer

The data layer contains:

- Government scheme database
- User profiles
- Scheme eligibility information
- Required documents
- Vector embeddings
- Conversation-related information

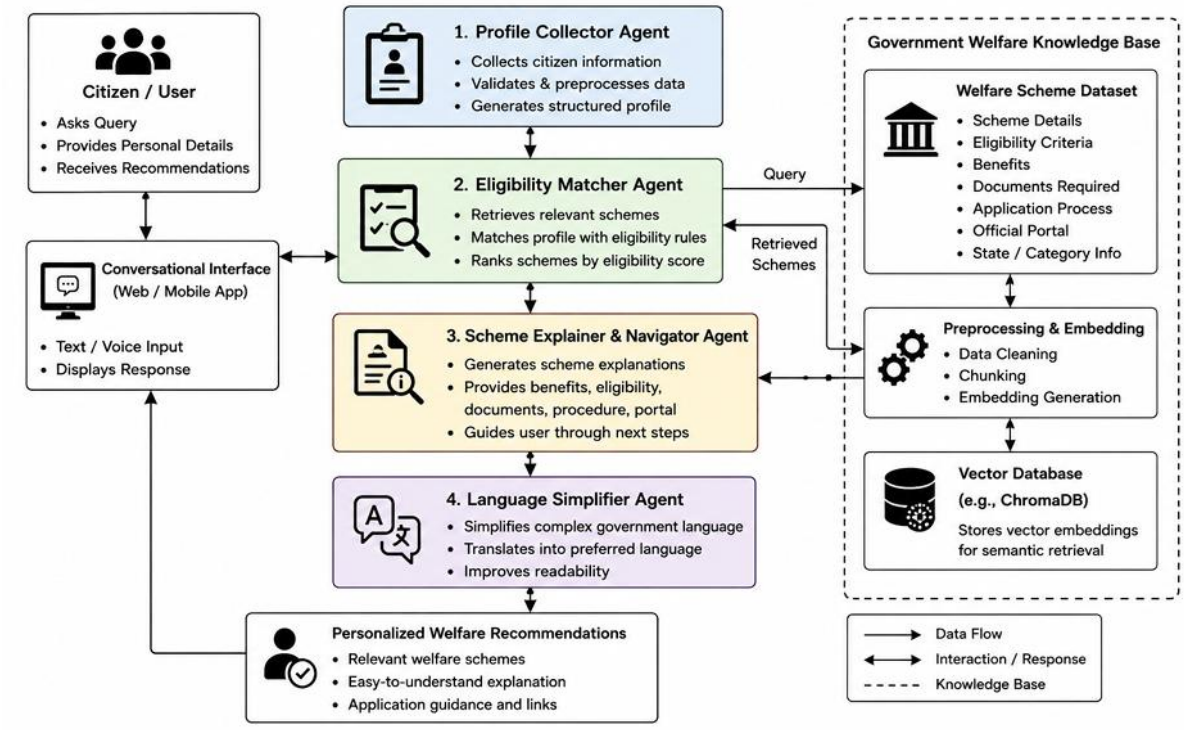
Table II

Technology Stack

Component	**Technology**
Frontend	React.js
UI Styling	Tailwind CSS
Backend	FastAPI
Programming Language	Python
Agent Orchestration	LangGraph
LLM/RAG Framework	LangChain
Embedding Model	Sentence Transformers

Component	Technology
Vector Database	ChromaDB
Relational Database	PostgreSQL
Data Format	JSON / CSV
Communication	REST APIs

The React.js frontend communicates with FastAPI backend services through REST APIs. LangGraph manages the flow between specialized agents, while LangChain supports the retrieval and LLM pipeline.



DATASET AND KNOWLEDGE BASE

The system uses a structured government welfare dataset containing **4,000+ central and state-level schemes**. The dataset contains information required for searching, eligibility assessment, recommendation, and explanation. The base paper similarly uses a structured dataset of more than 4,000 government schemes represented using JSON and normalized attributes.

Table III

Major Dataset Attributes

Attribute	Description
Scheme ID	Unique scheme identifier

Attribute	Description
Scheme Name	Official scheme name
Category	Agriculture, education, health, etc.
Beneficiary Group	Intended beneficiaries
Income Criteria	Applicable income conditions
Age Criteria	Applicable age requirements
Gender	Gender-specific eligibility
Occupation	Applicable occupation
State	Geographic applicability
Benefits	Benefits provided
Documents	Required documents
Application Portal	Official application information

Before retrieval, the dataset is cleaned by handling missing values, removing duplicates, normalizing textual fields, and generating semantic embeddings.

IMPLEMENTATION AND EXPERIMENTAL EVALUATION

The BenefitBridge AI system was implemented using the proposed multi-agent and RAG architecture. The frontend provides the citizen interaction interface, while FastAPI handles backend services and communication with the AI components.

The workflow begins when a citizen submits a query or profile. The Profile Collector Agent structures the information, after which the Eligibility Matcher Agent retrieves and evaluates relevant schemes. The Scheme Explainer Agent generates detailed scheme information, and the Language Simplifier Agent produces the final response in an accessible form.

A. Evaluation Metrics

The system is evaluated using recommendation quality, retrieval performance, response efficiency, and user interaction metrics.

Table IV

Evaluation Metrics

Metric	Purpose
Accuracy	Measures overall recommendation correctness
Precision	Measures relevance of recommended schemes
Recall	Measures eligible schemes successfully retrieved
F1-Score	Measures balance between precision and recall

Metric	Purpose
Response Time	Measures system response efficiency
Task Completion Rate	Measures successful completion of user interactions
User Satisfaction	Measures perceived usefulness of the system

B. Experimental Results

Table V

Performance Evaluation

Metric	Result
Accuracy	**95.8%**
Precision	**94.6%**
Recall	**93.9%**
F1-Score	**94.2%**
Average Response Time	**1.37 sec**
Task Completion Rate	**96.8%**
User Satisfaction	**4.7 / 5**

The results indicate that BenefitBridge AI provides accurate and relevant welfare scheme recommendations while maintaining low response latency. The high recall demonstrates that the system is capable of retrieving a large proportion of applicable schemes, while the precision value indicates that most recommended schemes are relevant to the user's profile.

The results also compare favorably with the reference approach, which reported approximately 92% eligibility matching accuracy, 90% Top-1 recommendation precision, and approximately 85% average recommendation precision.

AGENT-WISE PERFORMANCE

The execution time of each agent was measured to understand the contribution of individual components.

Table VI

Agent-Wise Execution Time

Agent	Average Execution Time
Profile Collector Agent	180 ms
Eligibility Matcher Agent	420 ms
Scheme Explainer & Navigator Agent	510 ms

Agent	Average Execution Time
Language Simplifier Agent	260 ms
Overall Pipeline	**1370 ms**

The Scheme Explainer and Navigator Agent requires comparatively more processing time because it retrieves relevant scheme information and generates detailed contextual responses. Despite this, the complete pipeline maintains an average response time of approximately 1.37 seconds.

OBSERVATIONS

The following observations were obtained from the experimental evaluation:

1. The system successfully identifies welfare schemes based on individual citizen profiles.
2. Personalized ranking reduces the need for citizens to manually search through large numbers of schemes.
3. RAG-based retrieval improves the grounding of generated responses in scheme-specific information.
4. The multi-agent architecture separates profile collection, eligibility analysis, explanation, and language simplification into independent tasks.
5. The conversational interface makes scheme information easier to understand.
6. Language simplification improves accessibility for citizens who may find government terminology difficult to interpret.
7. The modular architecture allows individual agents and knowledge components to be updated independently.

CONCLUSION

BenefitBridge AI provides an intelligent approach for simplifying access to government welfare schemes by combining **multi-agent AI, Retrieval-Augmented Generation, eligibility matching, and personalized recommendation**.

The system uses four specialized agents to collect citizen information, identify suitable schemes, explain recommendations, and simplify government terminology. Its integration with a structured welfare knowledge base enables responses to remain grounded in relevant scheme information.

Experimental evaluation achieved **95.8% accuracy, 94.6% precision, 93.9% recall, and 94.2% F1-score**, with an average response time of **1.37 seconds** and a task completion rate of **96.8%**. These results demonstrate the effectiveness of the framework in providing fast and personalized welfare assistance.

Overall, BenefitBridge AI acts as a bridge between citizens and government welfare services by reducing information overload, improving scheme discovery, and making complex government information easier to understand. The modular architecture also provides a strong foundation for expanding the system to additional government services and languages.